

Tim Perr

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Profile

Graduate student interested in computing and exploring the intersection of computing with various fields including science, finance, education, and the medical field. I like leveraging different computing tools such as Web Development, Data Science, AI, and Computing Theory to solve different problems that may or may not be related to computing.

Education

Michigan Technological University

MS Computer Science	Aug 2024–Dec 2025
MS in Data Science	Aug 2024–Dec 2025
Graduate Certificate Data Science	Aug 2024–Dec 2025
Graduate Certificate Applied Statistics	Aug 2024–Dec 2025
BS Computer Science	Aug 2021–Apr 2024

Skills

Programming Languages: Python, C/C++, Java, Swift, R, Matlab, Bash, MIPS Assembly, Makefile, LaTeX

Web Development: React.js, Node.js, HTML, CSS, JavaScript, Flask, PHP, APIs

Data and Machine Learning: SQL, Data Analysis, Machine Learning, Various Python libraries (e.g., NumPy, Pandas, Matplotlib), MATLAB Toolkits

Systems and Theoretical Knowledge: Concurrent/Parallel Computing, Quantum Computing, Data Structures, Programming Language Design, Computing Theory

Mathematics: Linear Algebra, Calculus, Statistics, Probability, Discrete Mathematics, Graph Theory

Artificial Intelligence: Various AI methods, AI in Healthcare, AI in Climate Science, Computer Vision, Fuzzy Systems, Genetic Algorithms

Other Related Skills: Linux, Git, Agile, basic Cyber Skills, Docker

Work and Research Experience

Computer Science Research Assistant

Michigan Technological University Feb 2023 – July 2025

- Led the development team of a website for Illuminated Devices, supported by an NSF-funded project [#BCS-2122034](#)
- Presented research on the theoretical aspects of the project at ASEE 2024; view the paper <https://peer.asee.org/46774>

Spinel Lab Member

Michigan Technological University Nov 2024 – Present

- Focused on Computer Science Education, specifically in reframing perceptions of computing and developing supplemental learning materials
- Responsible for data analysis and rewriting homework questions to enhance learning materials

Computer Science Learning Center Coach

Michigan Technological University Sep 2022 – Apr 2024

- Guided students through coursework across various levels, from introductory to graduate computing courses

Instructor

iD Tech Camps

May 2022 – Aug 2022

- Delivered curriculum to classes of up to 12 students, ensuring engagement and understanding of programming concepts
- Monitored students' progress toward learning goals and facilitated non-instructional activities

Bank Teller

First National Bank of LaGrange

Oct 2018 – Aug 2021

- Processed cash transactions, balanced drawers and vaults, issued cashier's checks, and delivered exceptional customer service

Projects

Illuminated Devices

Objective: Create a digital portal for remote digital assistance

- Patrons go to the local library to check out device, join a waiting queue when ready, then once a Tutor is ready they are connected for assistance
- Front end work done using React.js
- Back end work done using the python Flask framework utilizing a MariaDB database and SocketIO sockets
- Implemented the Zoom API to connect Patrons to the Tutors
- Two publications resulted from this work
- <https://github.com/tperr/illuminated-devices>

Brightness Lowerer

Objective: Make a tool that lowers brightness when rapid luminosity changes occur

- Programmed in python, with some bash interactions
- Grabs image of the screen, goes pixel by pixel to calculate the average luminosity
- Keeps average luminosity to a relatively consistent range
- Mostly a gimmick project, but fun and random idea
- https://github.com/tperr/brightness_lowerer

OpenFinAL

Objective: Create an open source financial analytics tool

- Programmed as coursework for BA5200 Information Systems Management with React.js, designed with Figma,
- An open source financial analytics tool designed to provide financial analytics capabilities at a low cost.
- It aims to help users understand their financial situation better and make informed decisions.
- My main focus was in the risk management module, which provides users with insights into the risk associated with their investments.
- <https://github.com/jeffreywallphd/OpenFinAL>